

7 In this question you must show detailed reasoning.

The complex number $\sqrt{3} + i$ is denoted by z .

- (a)** By expanding $(\sqrt{3} + i)^5$, express z^5 in the form $a + bi$ where a and b are real and exact. **[3]**
- (b) (i)** Express z in modulus-argument form. **[3]**
- (ii)** Hence find z^5 in modulus-argument form. **[2]**
- (iii)** Use this result to verify your answers to part **(a)** **[2]**